INCH-POUND

MIL-C-19978/9B
3 April 1995
SUPERSEDING
MIL-C-19978/9A
23 October 1973

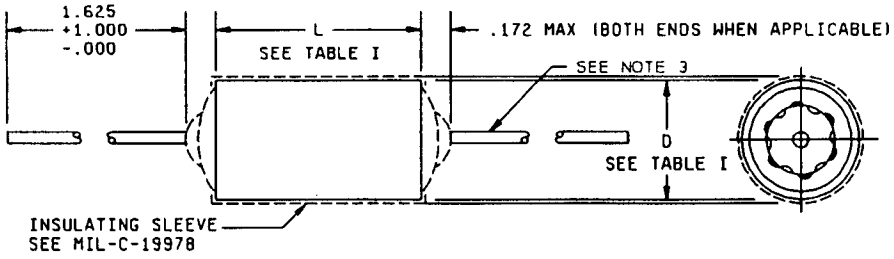
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR09

FAILURE RATE LEVEL 'L' IS INACTIVE FOR DESIGN AFTER 14 September 1983

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-19978



Inches	mm	Inches	mm
.001	0.03	.312	7.92
.004	0.10	.400	10.16
.025	0.64	.562	14.27
.032	0.81	1.000	25.40
.172	4.37	1.562	39.67
.235	5.97	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire, .025 (No. 22 AWG) for cases .235 and .312 in diameter; and .032 (No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as a wrap-around band. The supporting device will not be supplied with the capacitor.
5. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQR09 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

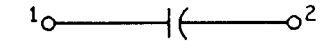
Part or Identifying Number (PIN): See table I.

Case material: Nonmagnetic (end seal may be of magnetic material).

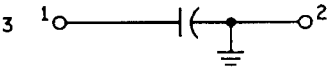
Dielectric material: Paper and polyethylene terephthalate (K).

Terminals: Axial-wire lead (see figure 1).

CIRCUIT DIAGRAM - 1



3



Rated voltage: See table I.

Rated temperature: -65°C to +125°C.

Capacitance (Cap.Δ) (nom): See table I.

Capacitance tolerance: See table I.

Dissipation factor (DF) (max): 1.0 percent.

Failure rate level: In accordance with MIL-C-19978.

Reliability: In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

TABLE I. Style CQR09 capacitors.

PIN 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/			
					Circuit 1		Circuit 3	
					L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	uF			Inches	Inches	Inches	Inches
CQR09A-KC392-3-	200	.0039	G,J,K	M,P,R,S	.750	.235	.688	.235
CQR09A-KC472-3-	200	.0047	G,J,K	"	"	"	"	"
CQR09A-KC562-3-	200	.0056	G,J,K	"	"	"	"	"
CQR09A-KC682-3-	200	.0068	G,J,K	"	"	"	"	"
CQR09A-KC183-3-	200	.018	G,J,K	"	.875	.312	.812	.312
CQR09A-KC223-3-	200	.022	G,J,K	"	"	"	"	"
CQR09A-KC273-3-	200	.027	G,J,K	"	"	"	"	"
CQR09A-KC333-3-	200	.033	G,J,K	"	"	"	"	"
CQR09A-KC393-3-	200	.039	G,J,K	"	.875	.400	.812	.400
CQR09A-KC473-3-	200	.047	G,J,K	"	"	"	"	"
CQR09A-KC563-3-	200	.056	G,J,K	"	"		"	"
CQR09A-KC683-3-	200	.068	G,J,K	"	"	"	"	"
CQR09A-KC823-3-	200	.082	G,J,K	"	1.125	.400	1.062	.400
CQR09A-KC104-3-	200	.10	G,J,K	"	"	"	"	"
CQR09A-KC124-3-	200	.12	G,J,K	"	1.375	.400	1.312	.400
CQR09A-KC154-3-	200	.15	G,J,K	"	"	"	"	"
CQR09A-KC184-3-	200	.18	G,J,K	"	1.125	.562	1.062	.562
CQR09A-KC224-3-	200	.22	G,J,K	"	"	"	"	"
CQR09A-KC274-3-	200	.27	G,J,K	"	1.375	.562	1.312	.562
CQR09A-KC334-3-	200	.33	G,J,K	"	"	"	"	"
CQR09A-KC394-3-	200	.39	G,J,K	"	1.625	.562	1.562	.562
CQR09A-KC474-3-	200	.47	G,J,K	"	"	"	"	"
CQR09A-KC564-3-	200	.56	G,J,K	"	1.625	.670	1.562	.670
CQR09A-KC684-3-	200	.68	G,J,K	"	"	"	"	"
CQR09A-KC824-3-	200	.82	G,J,K	"	2.125	.750	2.062	.750
CQR09A-KC105-3-	200	1.000	G,J,K	"	"	"	"	"
CQR09A-KE272-3-	400	.0027	G,J,K	"	.750	.235	.688	.235
CQR09A-KE332-3-	400	.0033	G,J,K	"	"	"	"	"
CQR09A-KE123-3-	400	.012	G,J,K	"	.875	.312	.812	.312
CQR09A-KE153-3-	400	.015	G,J,K	"	"	"	"	"
CQR09A-KE273-3-	400	.027	G,J,K	"	.875	.400	.812	.400
CQR09A-KE333-3-	400	.033	G,J,K	"	"	"	"	"
CQR09A-KE393-3-	400	.039	G,J,K	"	1.125	.400	1.062	.400
CQR09A-KE473-3-	400	.047	G,J,K	"	"	"	"	"
CQR09A-KE563-3-	400	.056	G,J,K	"	1.375	.400	1.312	.400
CQR09A-KE683-3-	400	.068	G,J,K	"	"	"	"	"
CQR09A-KE823-3-	400	.082	G,J,K	"	1.125	.562	1.062	.562
CQR09A-KE104-3-	400	.10	G,J,K	"	"	"	"	"
CQR09A-KE124-3-	400	.12	G,J,K	"	1.375	.562	1.312	.562
CQRo9A-KEl54-3-	400	.15	G,J,K	"	"	"	"	"
CQR09A-KE184-3-	400	.18	G,J,K	"	1.625	.562	1.562	.562
CQRo9A-KE224-3-	400	.22	G,J,K	"	"	"	"	"
CQR09A-KE274-3-	400	.27	G,J,K	"	1.625	.670	1.562	.670
CQR09A-KE334-3-	400	.33	G,J,K	"	"	"	"	"
CQR09A-KE394-3-	400	.39	G,J,K	"	2.125	.750	2.062	.750
CQR09A-KE474-3-	400	.47	G,J,K	"	"	"	"	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

TABLE I. Style CQR09 capacitors - Continued.

PIN 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/			
					Circuit 1		Circuit 3	
					L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	uF			Inches	Inches	Inches	Inches
CQR09A-KF102-3-	600	.0010	G,J,K	M,P,R,S	.750	.235	.688	.235
CQR09A-KF122-3-	600	.0012	G,J,K	"	"	"	"	"
CQR09A-KF152-3-	600	.0015	G,J,K	"	"	"	"	"
CQR09A-KF182-3-	600	.0018	G,J,K	"	"	"	"	"
CQR09A-KF222-3-	600	.0022	G,J,K	"	"	"	"	"
CQR09A-KF272-3-	600	.0027	G,J,K	"	.875	.312	.812	.312
CQR09A-KF332-3-	600	.0033	G,J,K	"	"	"	"	"
CQR09A-KF392-3-	600	.0039	G,J,K	"	"	"	"	"
CQR09A-KF472-3-	600	.0047	G,J,K	"	"	"	"	"
CQR09A-KF562-3-	600	.0056	G,J,K	"	"	"	"	"
CQR09A-KF682-3-	600	.0068	G,J,K	"	"	"	"	"
CQR09A-KF822-3-	600	.0082	G,J,K	"	"	"	"	"
CQR09A-KF103-3-	600	.010	G,J,K	"	"	"	"	"
CQR09A-KF123-3-	600	.012	G,J,K	"	.875	.400	.812	.400
CQR09A-KF153-3-	600	.015	G,J,K	"	"	"	"	"
CQR09A-KF183-3-	600	.018	G,J,K	"	"	"	"	"
CQR09A-KF223-3-	600	.022	G,J,K	"	"	"	"	"
CQR09A-KF273-3-	600	.027	G,J,K	"	1.125	"	1.062	"
CQR09A-KF333-3-	600	.033	G,J,K	"	"	"	"	"
CQR09A-KF393-3-	600	.039	G,J,K	"	1.375	.400	1.312	.400
CQR09A-KF473-3-	600	.047	G,J,K	"	"	"	"	"
CQR09A-KF563-3-	600	.056	G,J,K	"	1.125	.562	1.062	.562
CQR09A-KF683-3-	600	.068	G,J,K	"	"	"	"	"
CQR09A-KF823-3-	600	.082	G,J,K	"	1.375	.562	1.312	.562
CQR09A-KF104-3-	600	.10	G,J,K	"	"	"	"	"
CQR09A-KF124-3-	600	.12	G,J,K	"	1.625	.562	1.562	.562
CQR09A-KF154-3-	600	.15	G,J,K	"	"	"	"	"
CQR09A-KF184-3-	600	.18	G,J,K	"	1.625	.670	1.562	.670
CQR09A-KF224-3-	600	.22	G,J,K	"	"	"	"	"
CQR09A-KF274-3-	600	.27	G,J,K	"	2.125	.750	2.062	.750
CQR09A-KF334-3-	600	.33	G,J,K	"	"	"	"	"
CQR09A-KF394-3-	600	.39	G,J,K	"	2.375	.750	2.312	.750
CQR09A-KF474-3-	600	.47	G,J,K	"	"	"	"	"
CQR09A-KG102-3-	1000	.0010	G,J,K	"	.875	.400	.812	.400
CQR09A-KG122-3-	1000	.0012	G,J,K	"	"	"	"	"
CQR09A-KG152-3-	1000	.0015	G,J,K	"	"	"	"	"
CQR09A-KG182-3-	1000	.0018	G,J,K	"	"	"	"	"
CQR09A-KG222-3-	1000	.0022	G,J,K	"	"	"	"	"
CQRo9A-KG272-3-	1000	.0027	G,J,K	"	"	"	"	"
CQR09A-KG332-3-	1000	.0033	G,J,K	"	"	"	"	"
CQRo9A-KG392-3-	1000	.0039	G,J,K	"	"	"	"	"
CQR09A-KG472-3-	1000	.0047	G,J,K	"	"	"	"	"
CQR09A-KG562-3-	1000	.0056	G,J,K	"	"	"	"	"
CQR09A-KG682-3-	1000	.0068	G,J,K	"	"	"	"	"
CQR09A-KG822-3-	1000	.0082	G,J,K	"	"	"	"	"
CQR09A-KG103-3-	1000	.010	G,J,K	"	"	"	"	"

See footnotes at end of table.

TABLE I. Style CQR09 capacitors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

PIN 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/			
					Circuit 1		Circuit 3	
					L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	uF			Inches	Inches	Inches	Inches
CQR09A-KG123-3-	1000	.012	G,J,K	M,P,R,S	1.125	.400	1.062	.400
CQR09A-KG153-3-	1000	.015	G,J,K	"	"	"	"	"
CQR09A-KG183-3-	1000	.018	G,J,K	"	1.375	.400	1.312	.400
CQR09A-KG223-3-	1000	.022	G,J,K	"	"	"	"	"
CQR09A-KG273-3-	1000	.027	G,J,K	"	1.125	.562	1.062	.562
CQR09A-KG333-3-	1000	.033	G,J,K	"	"	"	"	"
CQR09A-KG393-3-	1000	.039	G,J,K	"	1.375	.562	1.312	.562
CQR09A-KG473-3-	1000	.047	G,J,K	"	"	"	"	"
CQR09A-KG563-3-	1000	.056	G,J,K	"	1.625	.562	1.562	.562
CQR09A-KG683-3-	1000	.068	G,J,K	"	"	"	"	"
CQR09A-KG823-3-	1000	.082	G,J,K	"	1.625	.670	1.562	.670
CQR09A-KG104-3-	1000	.10	G,J,K	"	"	"	"	"
CQR09A-KG124-3-	1000	.12	G,J,K	"	1.875	.670	1.812	.670
CQR09A-KG154-3-	1000	.15	G,J,K	"	"	"	"	"
CQR09A-KG184-3-	1000	.18	G,J,K	"	2.125	.750	2.062	.750
CQR09A-KG224-3-	1000	.22	G,J,K	"	"	"	"	"
CQR09A-KG274-3-	1000	.27	G,J,K	"	2.125	1.000	2.062	1.000
CQR09A-KG334-3-	1000	.33	G,J,K	"	"	"	"	"
CQR09A-KG394-3-	1000	.39	G,J,K	"	2.375	1.000	2.312	1.000
CQR09A-KG474-3-	1000	.47	G,J,K	"	"	"	"	"

1/ Complete PIN number shall include additional symbols to indicate circuit, capacitance tolerance, and failure rate level as applicable.

2/ Dimensions are for basic case; sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the capacitor body; however, if a shrink-fitted insulation is used for the sleeving, it shall lap over the ends of the capacitor body. Add .047 inch maximum to the nominal for capacitor diameter (for metric equivlanets of decimal inches, see table II).

TABLE II. Metric equivalents of decimal inches.

Inches	mm	Inches	mm
.005	0.13	0.812	20.62
.015	0.38	0.875	22.23
.016	0.41	1.000	25.40
.031	0.79	1.062	26.97
.047	1.19	1.125	28.58
.062	1.57	1.312	33.32
.235	5.97	1.375	34.93
.312	7.92	1.562	39.67
.400	10.16	1.625	41.28
.562	14.27	2.062	52.37
.670	17.02	2.125	53.98
.688	17.48	2.312	58.72
.750	19.05	2.375	60.33

Insulation resistance (IR):

Sleeving: In accordance with MIL-C-19978.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

Terminal to terminal: See table III.

TABLE III. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: Four hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (max): +10 percent.

Life:

\Capacitance change (max): ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS, SH, TD
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5910-1898)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:03Z
Check the source to verify that this is the current version before use.